

A Sensor based Secured Health Monitoring and Alert Technique using IoMT

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Abstract— A reliable health monitoring system essentially requires real time and continuous monitoring of critical health parameters of the patient and a prognostic approach to generate and send an alert during medical emergency. This paper proposes a sensor based health monitoring technique with adequate security features using Huffman Compression Coding and Rivest Shamir Adleman (RSA) encryption technique. Further, based on a prognostic approach an alert mechanism is developed with embedded software and internet of things. This proposed system is an improved version of existing systems and can be useful for patients on move or in intensive care unit.

Keywords—Alert system, Biomedical embedded system, Cloud Computing, Compression Coding, Data protection, Encryption, Health Monitoring System, IoT, IoMT, SMS.

I. INTRODUCTION

The monitoring of health parameters for person suffering from chronic disease or for patients under intensive care unit is very important in order to deal with medical emergencies. However, the timing of information sent to doctor or nursing staff plays an important role. Further, the security of data is a major concern in such type of system because it carries sensitive and crucial medical database. The transfer of information over an internet channels carrying crucial data pertaining to patient's health may be corrupted on account of multiple types of security threats. Therefore, a robust and secured system is essential with efficient and adequate encryption technique. Challenges for data protection includes: physical security, secure routing, data transparency, security of handling IoT data etc. Sensor based wearable electronics hardware should be preferred for ease of operation in patient's in move or work place or in intensive care unit. An alert mechanism based on prognostic approach is highly desirable and embedded software is developed in order to send alert message to medical authorities. The primary health parameters considered in proposed technique are ECG, body temperature, blood pressure, Galvanic skin response etc.

Internet of Things (IoT) is a technology which enables real time data gain, data transfer, device connectivity, to control end user application. With the quick development of web innovation and communication technology, our everyday schedules are progressively focused on virtual world [1]. The term IoMT(Internet of Medical Things), basically used in the healthcare application which consists of connected devices that sense vital data of human physiological parameter in real time. IoT is most promising or better way in healthcare applications since it causes

patients to deal with their very own illness and get help in most crisis case by means of portable devices [2].

A lot of research work has been done in this field. With the use of mobile computing [3], resultant output is displayed on android by interfacing microcontroller arduino with LM35 temperature sensor and pulse sensor. In [4] all the sensed data send to Red-Node platform through MQTT protocol. Through Node-Red, it uses the SMS and email services. The implementation [5] of two sensors is interfaced with arduino board and the resultant output is shown on output console. The patient condition's data are stored and obtained using IoT technology. GSM technology and LabView software [6] is used for analyzing the data and gives an alert to doctor using GSM900A. Various symmetric and asymmetric algorithm of encryption is discussed [7]. Comparison of the symmetric and asymmetric cryptography used in network security on the basis of theoretical performance analysis is discussed [8]. In above mentioned research and survey paper, there are some limitations of network security, time delay, major physiological health parameter and many more which are discussed and overcome in our proposed technique. In this paper, we have proposed and described a novel method of developing secured health monitoring and alert system based on IoMT.

II. SYSTEM MODEL

The system model for the proposed technique is shown in fig.1. This model consists of major three section viz. Hardware section, Software section and Cloud section. Sensing unit and receiver unit are the part of hardware section in which microcontroller acts as a bridge between the two.

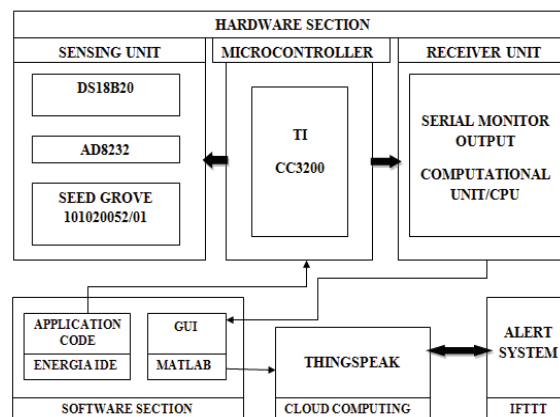


Fig. 1 Schematic diagram of proposed technique

IoT enabled TI CC3200 microcontroller is used, a high performance ARM CORTEX M4 controller [9] having built-in WiFi connectivity. It works on different power modes like active mode, sleep mode, Hibernate mode etc. thus power consumption is low as compared to arduino UNO.

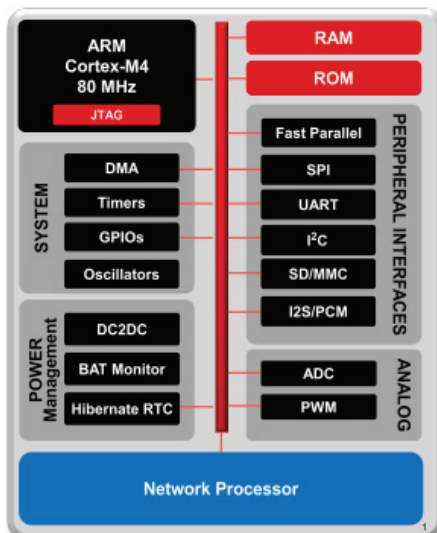


Fig. 2 CC3200 Hardware Overview [9]

Sensing unit consists body temperature sensor, ECG sensor and Galvanic skin response sensor. Body temperature is easy to observe and it can depict various health issue. Ds18b20 is a digital temperature sensor with one wired configuration and having a range of -55 to +125 °C with accuracy +/-0.5 °C. It has a 3 connection. Red for power supply, black for ground and yellow is a data line where we get output signal.

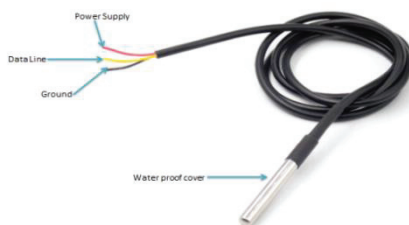


Fig. 3 DS18B20 Temperature Sensor

Galvanic skin response sensor is used to measure skin conductivity with the help of resistance of a body. As the sweating increases, resistance decreases thus conductivity increases as per inverse property.

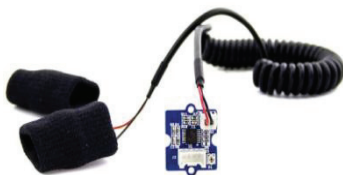


Fig. 4 GSR Model number 101020052/01

It is having two probes to measure signals. Strong emotion activates sympathetic nervous system, cause more sweating. Thus it is used in medical world to find out emotion of patient such as fear, frightened, sock etc. with

the help sweating formula. It has 4 slot for connection in which white is for no connection, red is for 5V, Black is for ground and yellow is for output.

AD8232 ECG sensor is most widely used. It can measure the electrical activity of a heart. 5 pins out 9 pins are used for measurement, in which GND is for ground, LO⁻ and LO⁺ is for lead off detect – and + respectively and other two are for output and power supply.



Fig.5. AD8232 with probes

Receiver unit as shown in fig. 1 consists of computational unit viz. output screen where results are shown via sensing unit.

Software section contains only two parts viz. Energia IDE where application code is generated and Matlab is for data analysis. Energia software is an open source IDE (Integrated Development Environment) based on wiring and arduino framework. It is supported on Mac OS, Windows, and Linux. This platform was started in 2012 by Texas Instrumentation. TI offers a CC32xx, CC13xx, MSP430 etc. launchpad to use with Energia IDE. It is similar to arduino IDE. Matlab software is used which is a powerful software, allows analysis of data plotting of function, implementation of algorithm, matrix manipulation, GUI etc. and acts as a medium for cloud computing.

Cloud section consists of thingspeak cloud which is interlinked with alert mechanism process. Thingspeak cloud is an internet of thing platform used to collect and store and data. It allows analysis and visualization of data in MATLAB. It has many features in which we discuss on storage of our data, MATLAB analysis and generation of alert system.

III. PROPOSED SECURITY ALGORITHM

A. Huffman Compression Coding

While using IoT technologies, we have limited storage capacity to store or save patient's data. Storage capacity of cloud is also increasing day by day but instead of this, we also make sure that large amount of subject's data can be store in minimum place. Compression of the data is best option for this. It can be lossy or lossless compression of data. In medical field, lossless compression technique should be considered because medical data is very sensitive [10], minor change can cause many problems.

Huffman compression coding is also known as Huffman encoding and having a quality of lossless compression [11]. It is used for data compression. Huffman is worked on the

prefix code and binary tree rules. Character is generally stored in 8 bit in the series of “0” and “1” and this is called fixed length encoding. In Huffman, here in proposed technique variable length encoding depending upon the frequency of character viz. how many times that character appears. Thus storing the character on that length only which are needed for thus size of bits is reduced. Huffman encodes the data thus it changes the expression of data and act as a first step of security. Huffman is easy to implement and optimal. Thus it is preferred. The relevant performance parameters for compression are as follows:

Compression Ratio: It is a ratio between the original size file and compressed file size [12].

$$CR = \frac{\text{size of a original file}}{\text{size of a compressed file}} \quad (1)$$

Percentage root mean square difference:

$$PRD = \sqrt{\frac{\sum_{i=1}^n (\text{original signal} - \text{decompressed signal})^2}{\sum_{i=1}^n (\text{original signal})^2}} \quad (2)$$

Quality ratio: It is a difference between compression ratio and PRD.

$$QR = \frac{CR}{PRD} \quad (3)$$

B. Rivest Shamir Adleman Encryption

The protection of captured health information from different sensors furthermore from unlawful access is critical. Accordingly, stringent approaches and specialized safety efforts ought to be presented to impart patient’s information with approved users and organization. Challenges for data protection includes: physical security, secure routing, data transparency, security of handling lot data etc. One of the most used and secure algorithm is Rivest Shamir Adleman (RSA). RSA is one of the best cryptographic algorithms [13] in present use that ensures secure communication over networks because of its long key generation manner. We took the advantage of difficulty in factorization of large numbers because medical data is very crucial.

TABLE I ALGORITHM FOR RSA

Algorithm for RSA
• Compute two prime number i.e. a and b where a is not equal to b. • Calculate mod c i.e. $c = a * b$. • Calculate phi ϕ where $\phi = (a - 1) * (b - 1)$ • Calculate exponent e where $e \neq$ factor of c and satisfy equation $1 < e < \phi(c)$ • Obtain public key (c, e) and represent the plaintext message as a positive integer m. • Compute the cipher text $p = m^e \pmod{c}$. • Using private key (c, d) to compute $m = p^d \pmod{c}$. • Extract the plaintext from cipher text m.

IV. METHODOLOGY

The methodology to execute the proposed technique is indicated in flow diagram of fig. 6.

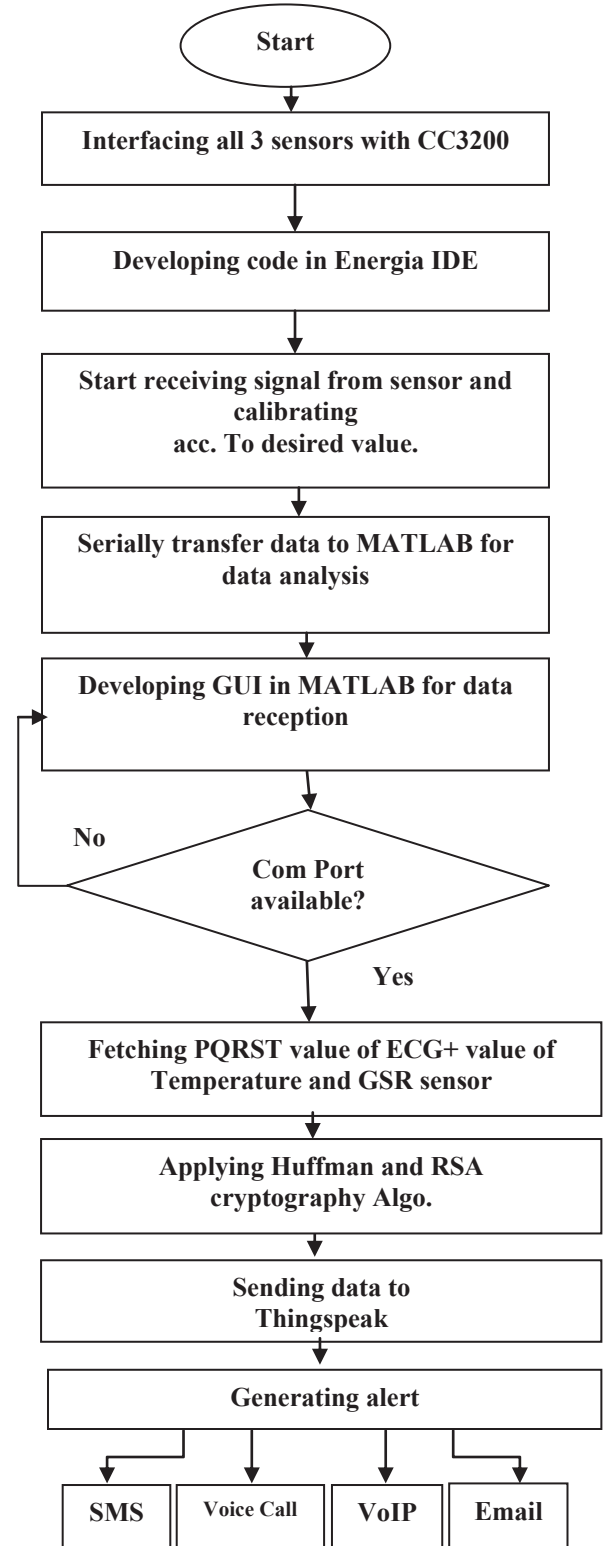


Fig.6 Flow Diagram for proposed technique

As per initial process of proposed techniques, there is interfacing of sensors (ECG, body temperature, GSR) with the controller CC3200 by the use of analog to digital converter. Further, code for sensor is developed in Energia

IDE. Receiver unit starts receiving signal and calibrating according to desired value.

Now, data is serially transferred to MATLAB for data analysis and graphical user interface (GUI) is developed for data reception. It checks the condition whether Com port is available or not. If it is available, then further process is processed and if not then it again checks for com port availability.

After connecting to com port fetching of all three sensor value is done. Now applying the Huffman Compression and RSA encryption algorithm on the received data to compress and secure the data. Thingspeak cloud is connected and all encrypted data is sent to cloud with the help of WiFi connectivity.

Alert mechanism is generated with the help of cloud and IFTTT (If That This That). The whole process can be repeated for the next packet of signal if desired.

A. Alert Mechanism

Alert system is used when emergency occur in patient's health data. Here with the help of features of Thingspeak apps an alert message can be generated in four ways which are as follows:-

1. Alert Through SMS
2. Alert Through Voice Call
3. Alert Through VoIP(Voice over Internet Protocol)
4. Alert Through Email.

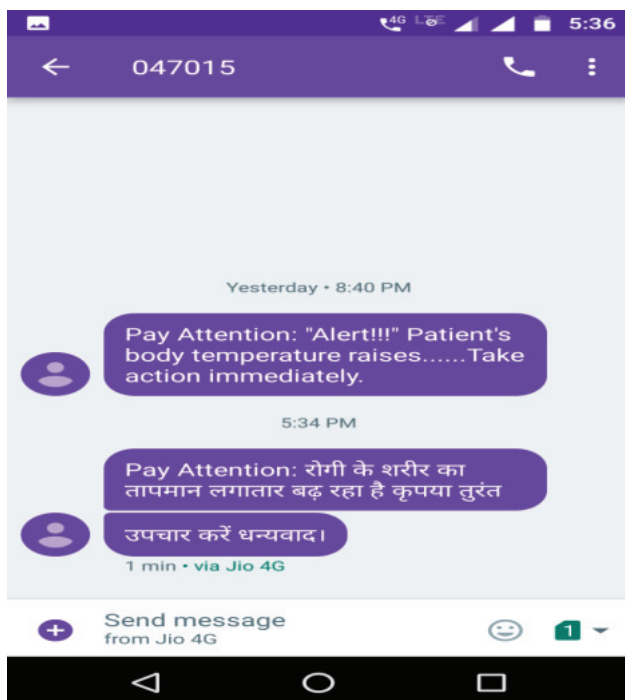


Fig. 7: Screenshot of SMS alert

Depending on the specific condition, priorities can be set for each alert. Alert through SMS enables a short message to be received by a doctor in any language. SMS is a best way to receive alert because it a permanent message unless or until we delete. Alert through mail is also works like SMS,

an email is sent to specific person or multiple people. Alert through VoIP and Voice Call is same but having only one difference i.e. Voice Call is a direct call and VoIP is made through app via internet connection.

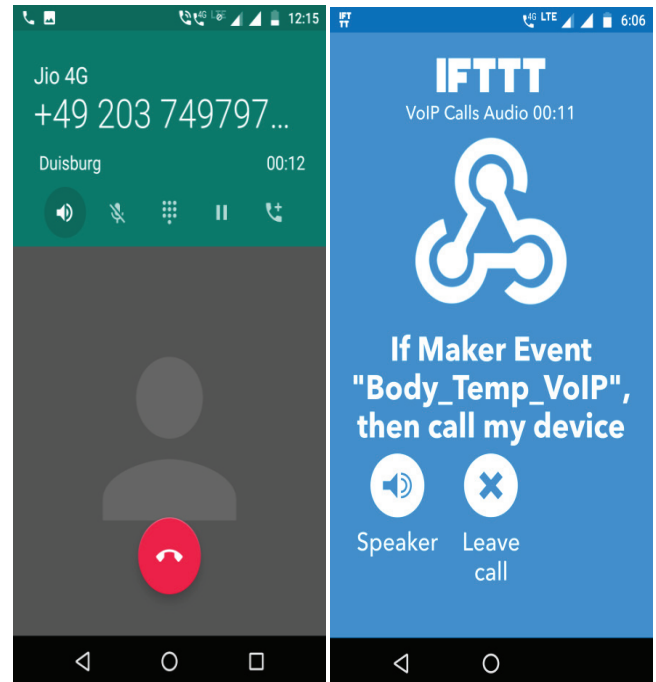


Fig. 8 Screenshot of voice call and VoIP

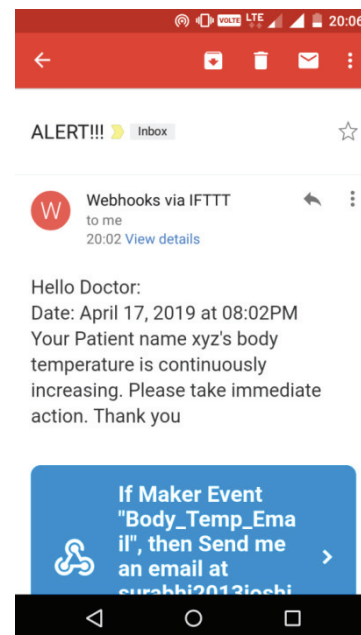


Fig. 9 Screenshot of Email alert

Algorithm for alert is developed for proposed technique. Generation of SMS alert is coded in two languages i.e. English and Hindi. Results are shown in the fig. 7, 8 and 9.

V. CONCLUSION

A novel sensor based secured health monitoring technique is proposed and discussed. Hybrid model of Huffman compression and RSA encryption provides two level of encoding which enhance the data protection thus they are preferred. Outcomes of this proposed technique is

expected to be effective because important and sensible physiological parameter is used as they are considering as the roots of basic normal parameter of health. Secondly, compression technique increases the storage capacity with low time delay. Thirdly, encryption algorithm secures the network path and without decryption key no one can reformed the actual data. Lastly, using four alert systems we will have a four alternate way to give message about emergency occurrence. So by combining data compression and encryption algorithm provides a very strong result in terms of speed, storage capacity, network security and data protection. Four alternate alert systems provide magnified and excellent performance in emergency. Existing system suffers from compression of data with data protection technique and having none or single alert mechanism. Thus as a conclusion this proposed techniques will be fruitful for the design of health monitoring system.

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